

2021 AIME II Problems/Problem 5

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Problem

For positive real numbers s , let $\mathcal{T}(s)$ denote the set of all obtuse triangles that have area s and two sides with lengths 4 and 10. The set of all s for which $\mathcal{T}(s)$ is nonempty, but all triangles in $\mathcal{T}(s)$ are congruent, is an interval $[a, b]$. Find $a^2 + b^2$.

Solution 1

We start by defining a triangle. The two small sides MUST add to a larger sum than the long side. We are given 4 and 10 as the sides, so we know that the 3rd side is between 6 and 14, exclusive. We also have to consider the word OBTUSE triangles. That means that the two small sides squared is less than the 3rd side. So the triangles' sides are between 6 and $\sqrt{84}$ exclusive, and the larger bound is between $\sqrt{116}$ and 14, exclusive. The area of these triangles are from 0 (straight line) to $2\sqrt{84}$ on the first "small bound" and the larger bound is between 0 and 20. $0 < s < 2\sqrt{84}$ is our first equation, and $0 < s < 20$ is our 2nd equation. Therefore, the area is between $\sqrt{336}$ and $\sqrt{400}$, so our final answer is 736.

~ARCTICTURN

Solution 2 (Inequalities and Casework)

If a , b , and c are the side-lengths of an obtuse triangle with $a \leq b \leq c$, then both of the following must be satisfied:

- *Triangle Inequality Theorem:* $a + b > c$
- *Pythagorean Inequality Theorem:* $a^2 + b^2 < c^2$

For one such obtuse triangle, let 4, 10, and x be its side-lengths and K be its area. We apply casework to its longest side:

Case (1): The longest side has length 10, so $0 < x < 10$.

By the Triangle Inequality Theorem, we have $4 + x > 10$, from which $x > 6$.

By the Pythagorean Inequality Theorem, we have $4^2 + x^2 < 10^2$, from which $x < \sqrt{84}$.

Taking the intersection produces $6 < x < \sqrt{84}$ for this case.

At $x = 6$, the obtuse triangle degenerates into a straight line with area $K = 0$; at $x = \sqrt{84}$, the obtuse triangle degenerates into a right triangle with area $K = \frac{1}{2} \cdot 4 \cdot \sqrt{84} = 2\sqrt{84}$. Together, we obtain $0 < K < 2\sqrt{84}$, or $K \in (0, 2\sqrt{84})$.

Case (2): The longest side has length x , so $x \geq 10$.

By the Triangle Inequality Theorem, we have $4 + 10 > x$, from which $x < 14$.

By the Pythagorean Inequality Theorem, we have $4^2 + 10^2 < x^2$, from which $x > \sqrt{116}$.

Taking the intersection produces $\sqrt{116} < x < 14$ for this case.

At $x = 14$, the obtuse triangle degenerates into a straight line with area $K = 0$; at $x = \sqrt{116}$, the obtuse triangle degenerates into a right triangle with area $K = \frac{1}{2} \cdot 4 \cdot 10 = 20$. Together, we obtain $0 < K < 20$, or $K \in (0, 20)$.

Answer

It is possible for noncongruent obtuse triangles to have the same area. Therefore, all such positive real numbers s are in exactly one of $(0, 2\sqrt{84})$ or $(0, 20)$. By the exclusive disjunction, the set of all such s is

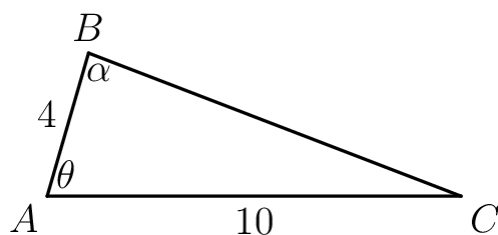
$$[a, b) = (0, 2\sqrt{84}) \oplus (0, 20) = [2\sqrt{84}, 20),$$

from which $a^2 + b^2 = \boxed{736}$.

~MRENTHUSIASM

Solution 3

We have the diagram below.



We proceed by taking cases on the angles that can be obtuse, and finding the ranges for s that they yield.

If angle θ is obtuse, then we have that $s \in (0, 20)$. This is because $s = 20$ is attained at $\theta = 90^\circ$, and the area of the triangle is strictly decreasing as θ increases beyond 90° . This can be observed from

$$s = \frac{1}{2}(4)(10) \sin \theta$$

by noting that $\sin \theta$ is decreasing in $\theta \in (90^\circ, 180^\circ)$.

Then, we note that if α is obtuse, we have $s \in (0, 4\sqrt{21})$. This is because we get

$x = \sqrt{10^2 - 4^2} = \sqrt{84} = 2\sqrt{21}$ when $\alpha = 90^\circ$, yielding $s = 4\sqrt{21}$. Then, s is decreasing as α increases by the same argument as before.

$\angle ACB$ cannot be obtuse since $AC > AB$.

Now we have the intervals $s \in (0, 20)$ and $s \in (0, 4\sqrt{21})$ for the cases where θ and α are obtuse, respectively. We are looking for the s that are in exactly one of these intervals, and because $4\sqrt{21} < 20$, the desired range is

$$s \in [4\sqrt{21}, 20)$$

giving

$$a^2 + b^2 = \boxed{736}.\square$$

Solution 4

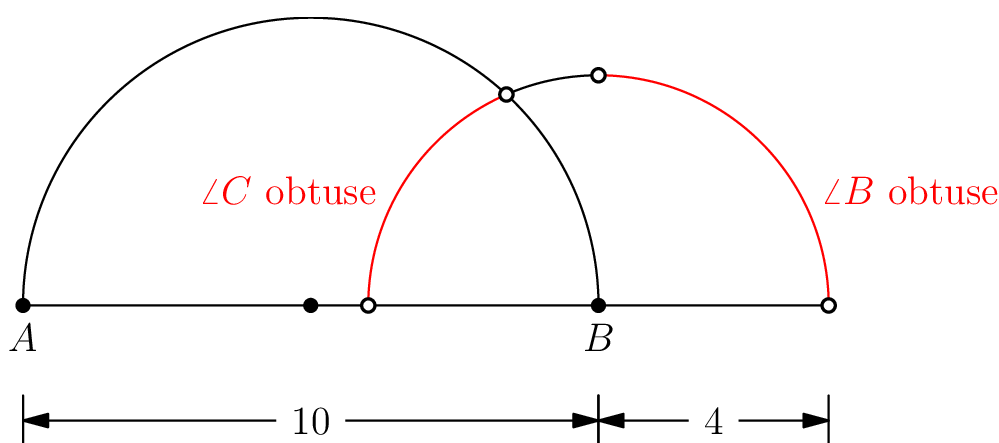
Note: Archimedes15 Solution which I added an answer here are two cases. Either the 4 and 10 are around an obtuse angle or the 4 and 10 are around an acute triangle. If they are around the obtuse angle, the area of that triangle is < 20 as we have $\frac{1}{2} \cdot 40 \cdot \sin \alpha$ and $\sin$ is at most 1. Note that for the other case, the side lengths around the obtuse angle must be 4 and x where we have $16 + x^2 < 100 \rightarrow x < 2\sqrt{21}$. Using the same logic as the other case, the area is at most $4\sqrt{21}$. Square and add $4\sqrt{21}$ and 20 to get the right answer

$$a^2 + b^2 = \boxed{736}.\square$$

Solution 5 (Circles)

For $\triangle ABC$, we fix $AB = 10$ and $BC = 4$. Without the loss of generality, we consider C on only one side of $\overline{AB}$.

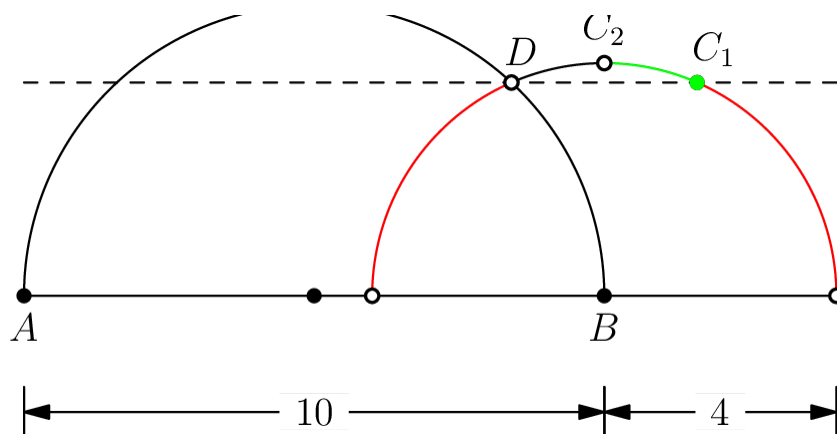
As shown below, all locations for C at which $\triangle ABC$ is an obtuse triangle are indicated in red, excluding the endpoints.



Note that:

1. The region in which $\angle B$ is obtuse is determined by construction.
2. The region in which $\angle C$ is obtuse is determined by the corollaries of the Inscribed Angle Theorem.

For any fixed value of s , the height from C is fixed. We need obtuse $\triangle ABC$ to be unique, so there can only be one possible location for C . As shown below, all possible locations for C are on minor arc $\widehat{C_1C_2}$, including C_1 but excluding C_2 .



Let the brackets denote areas:

- If $C = C_1$, then $[ABC]$ will be minimized (attainable). By the same base and height and the Inscribed Angle Theorem, we have

$$\begin{aligned}
 [ABC] &= [ABD] \\
 &= \frac{1}{2} \cdot BD \cdot DA \\
 &= \frac{1}{2} \cdot BD \cdot \sqrt{AB^2 - BD^2} \\
 &= \frac{1}{2} \cdot 4 \cdot \sqrt{10^2 - 4^2} \\
 &= 2\sqrt{84}.
 \end{aligned}$$

- If $C = C_2$, then $[ABC]$ will be maximized (unattainable). For this right triangle, we have

$$\begin{aligned}
 [ABC] &= \frac{1}{2} \cdot AB \cdot BC \\
 &= \frac{1}{2} \cdot 10 \cdot 4 \\
 &= 20.
 \end{aligned}$$

Finally, the set of all such s is $[a, b) = [2\sqrt{84}, 20)$, from which $a^2 + b^2 = \boxed{736}$.

~MRENTHUSIASM (credit given to Snowfan)

Solution 6

Let a triangle in $\tau(s)$ be ABC , where $AB = 4$ and $BC = 10$. We will proceed with two cases:

Case 1: $\angle ABC$ is obtuse. If $\angle ABC$ is obtuse, then, if we imagine AB as the base of our triangle, the height can be anything in the range $(0, 10)$; therefore, the area of the triangle will fall in the range of $(0, 20)$.

Case 2: $\angle BAC$ is obtuse. Then, if we imagine AB as the base of our triangle, the height can be anything in the range $(0, \sqrt{10^2 - 4^2})$. Therefore, the area of the triangle will fall in the range of $(0, 2\sqrt{84})$.

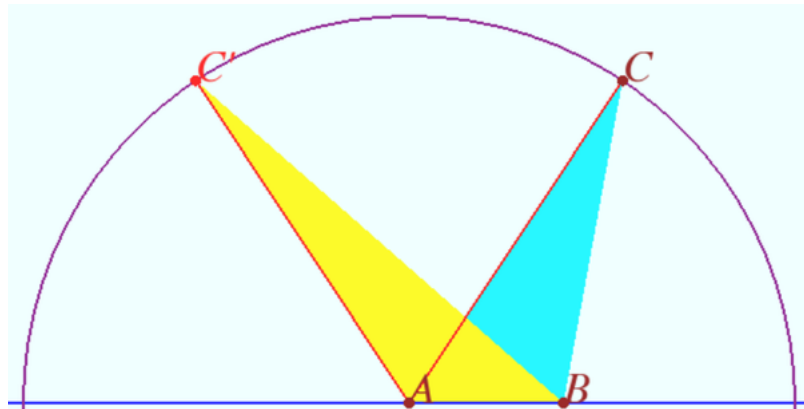
If $s < 2\sqrt{84}$, there will exist two types of triangles in $\tau(s)$ - one type with $\angle ABC$ obtuse; the other type with $\angle BAC$

obtuse. If $s \geq 2\sqrt{84}$, as we just found, $\angle BAC$ cannot be obtuse, so therefore, there is only one type of triangle - the one in which $\angle ABC$ is obtuse. Also, if $s > 20$, no triangle exists with lengths 4 and 10. Therefore, s is in the range $[2\sqrt{84}, 20)$, so our answer is $(2\sqrt{84})^2 + 20^2 = \boxed{736}$.

Alternatively, refer to Solution 5 for the geometric interpretation.

~ihatmath123

Solution 7



Let's rephrase the condition. It is required to find such values of the area of an obtuse triangle with sides 4 and 10, when there is exactly one such obtuse triangle. In the diagram, $AB = 4$, $AC = 10$.

The largest area of triangle with sides 4 and 10 is 20 for a right triangle with legs 4 and 10 ($AC \perp AB$).

The diagram shows triangles with equal heights. The yellow triangle ABC' has the longest side BC' , the blue triangle ABC has the longest side AC . If $BC \perp AB$, then $BC = \sqrt{AC^2 - AB^2} = 2\sqrt{21}$ the area is equal to $4\sqrt{21}$. In the interval, the blue triangle ABC is acute-angled, the yellow triangle ABC' is obtuse-angled. Their heights and areas are equal. The condition is met.

If the area is less than $4\sqrt{21}$, both triangles are obtuse, not equal, so the condition is not met.

Therefore, s is in the range $[4\sqrt{21}, 20)$, so answer is $(4\sqrt{21})^2 + 20^2 = \boxed{736}$

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Solution 8

If 4 and 10 are the shortest sides and $\angle C$ is the included angle, then the area is

$$\frac{4 \cdot 10 \cdot \sin \angle C}{2} = 20 \sin \angle C.$$

Because $0 \leq \sin \angle C \leq 1$, the maximum value of $20 \sin \angle C$ is 20, so $s \leq 20$.

If 4 is a shortest side and 10 is the longest side, the length of the other short side is

$4 \cos \angle C + 2\sqrt{4 \cos^2 \angle C + 21}$ by law of cosines, and the area is

$2 \left(4 \cos \angle C + 2\sqrt{4 \cos^2 \angle C + 21} \right) \sqrt{1 - \cos \angle C}$. Because $-1 \leq \cos \angle C \leq 0$, this is minimized if

$$\cos \angle C = 0, \text{ where } s = 4\sqrt{21}.$$

$$\text{So, the answer is } 20^2 + \left(4\sqrt{21}\right)^2 = \boxed{736}.$$

~ryanbear

Video Solution by Interstigation

<https://youtu.be/ODokTet3EVA>

~Interstigation

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